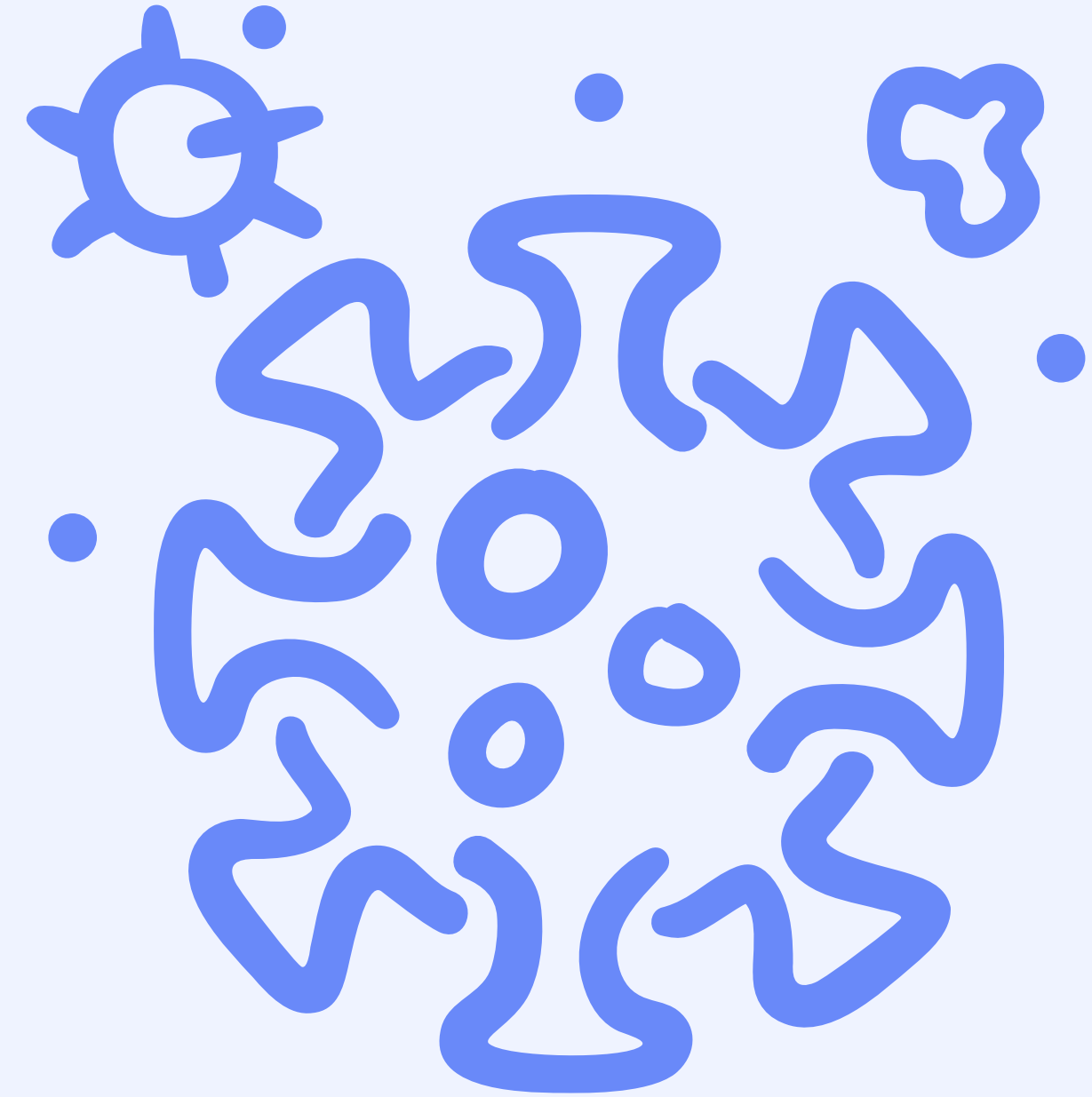


Vectorization for therapy

Application in Huntington's Disease

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OBONE ESSONO Heva
QUEZADA SOLIS Diana
VERDIER Adélie



Hungtintong's disease

Hungtintong's disease

Heredity and symptoms

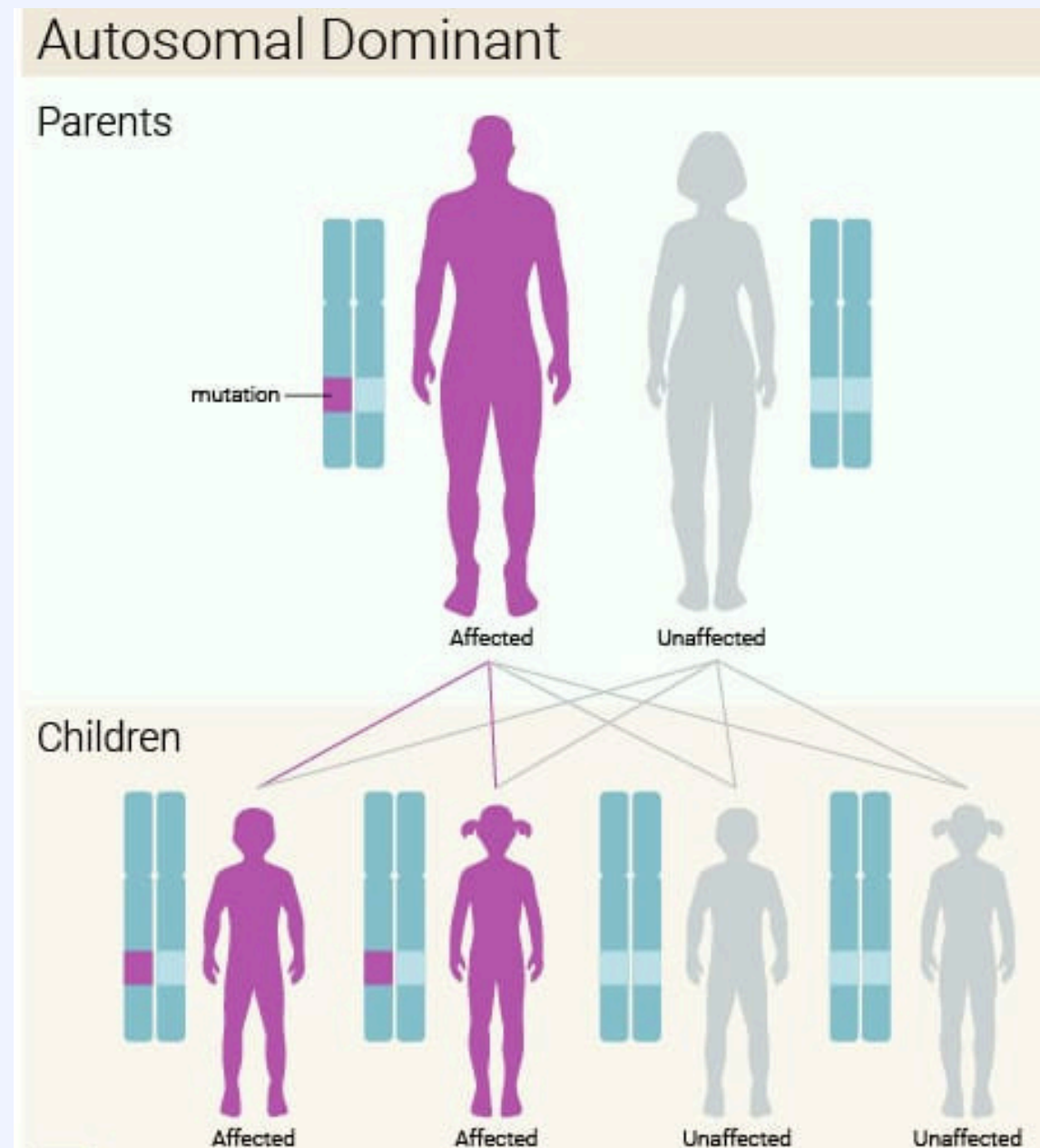


Figure 1: *HTT* gene inheritance [1].

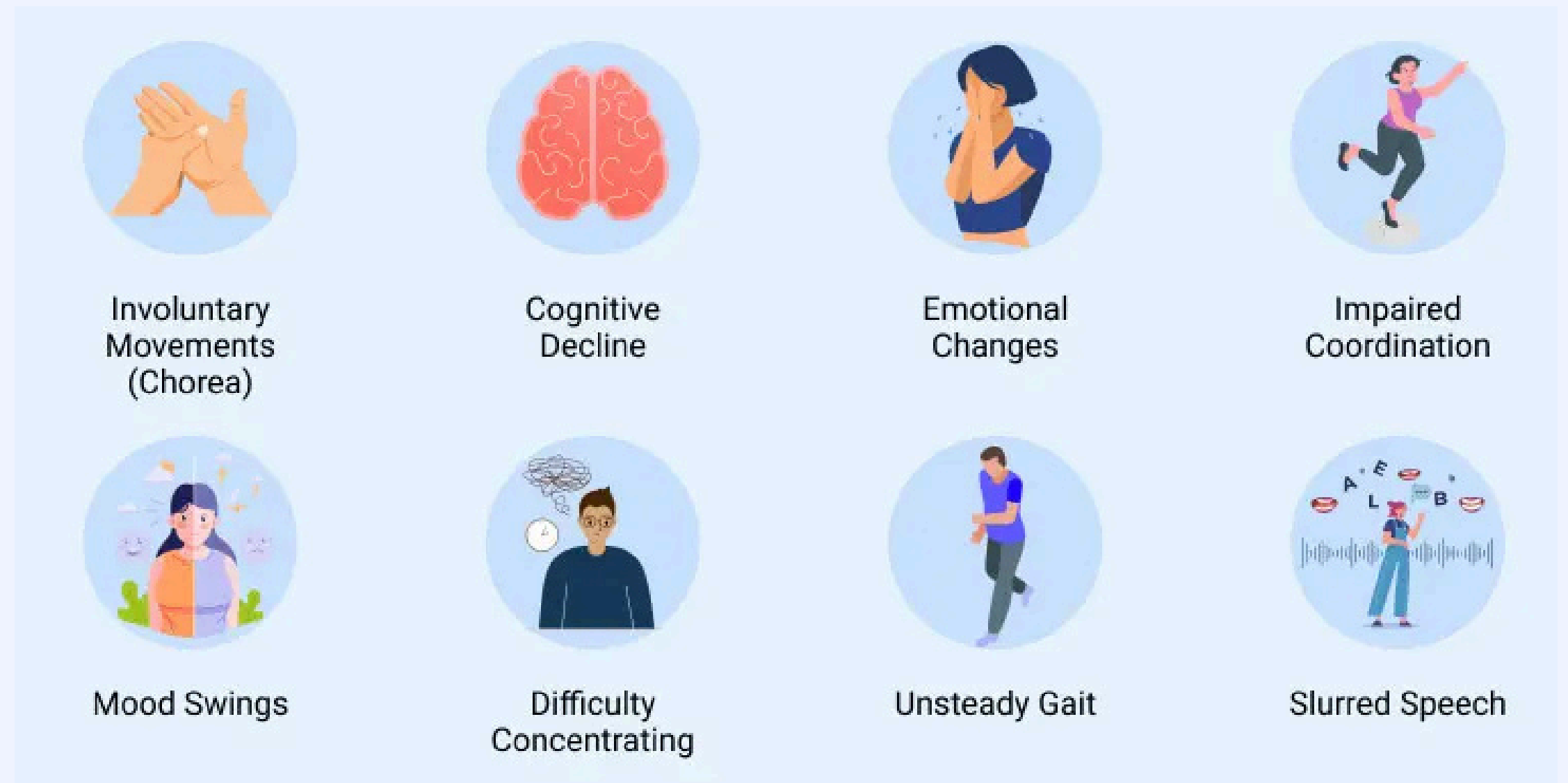
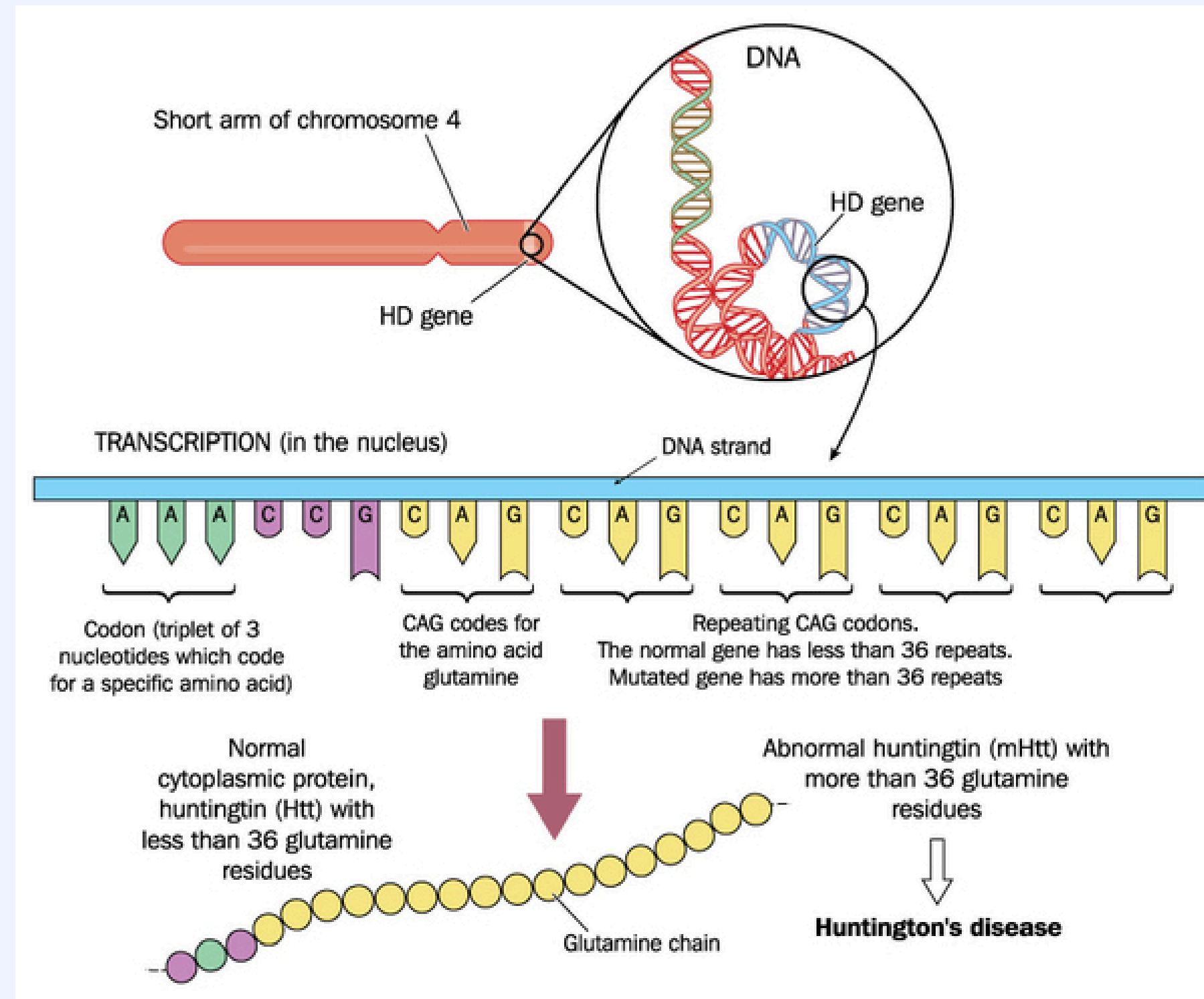


Figure 2: *Common symptoms of Huntington's Disease* [2].

Hungtintong's disease

Cause



Hungtintong's disease

Pathophysiology.

- Neuronal aggregates.
- Transcriptional dysregulation.
- Excitotoxicity.
- Mitochondrial dysfunction and altered.
- Energy metabolism.

Complexity of Treatments

Polymedication

MULTIPLICITY OF SYMPTOMS

MOLECULAR INTERACTIONS

SIDE EFFECTS

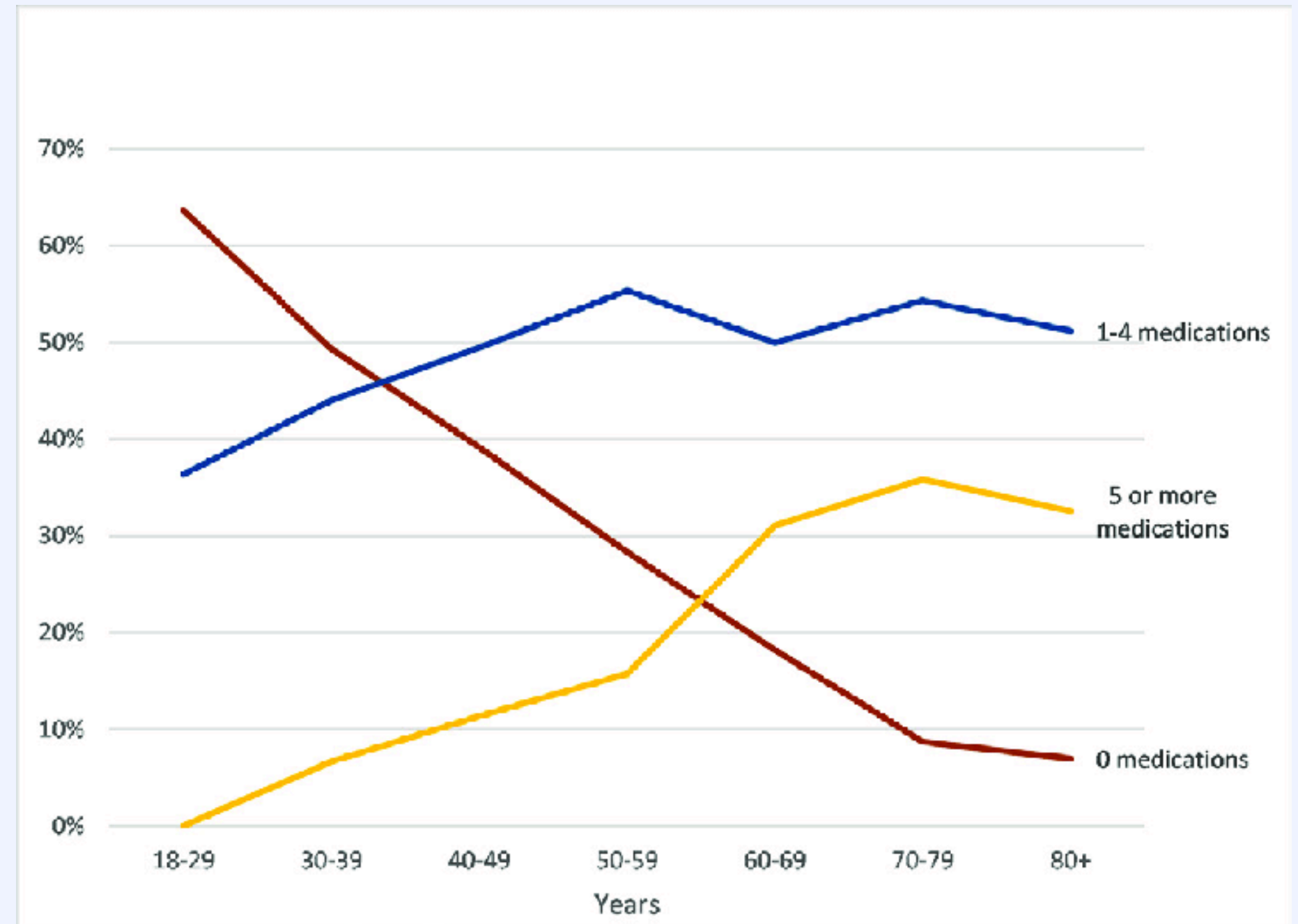


Figure 4: Relationship between age and medication number [4].

Complexity of Treatments

Biological Barrier and transportation

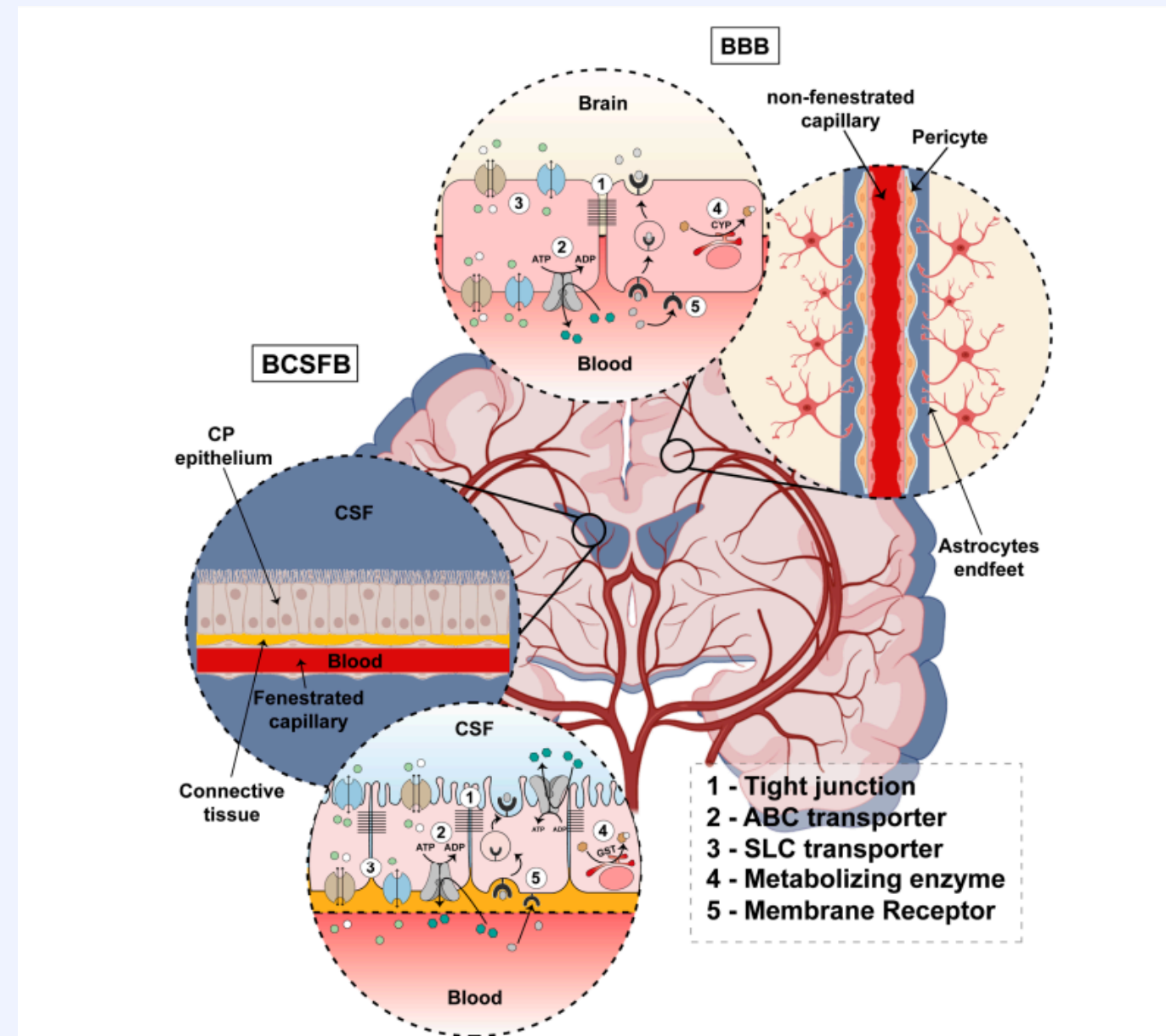


Figure 5: Schematic illustration of the blood–brain barrier (BBB) and blood–cerebrospinal fluid barrier (BCSFB) [5].

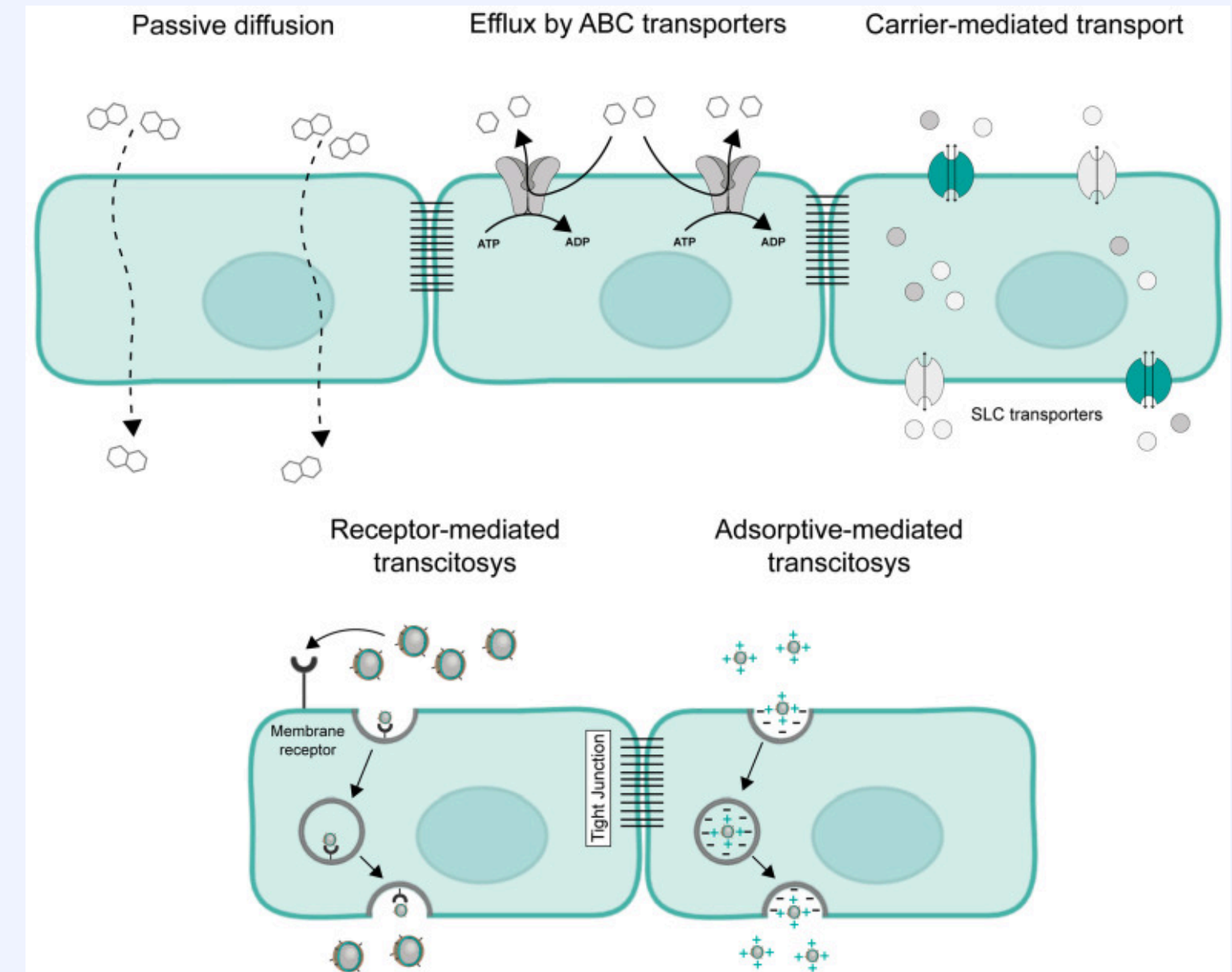


Figure 6: Mechanisms of transport across brain barriers [6].

The background features several light blue, wavy, double-lined curves that flow across the slide, framing the central text.

Vectorization

Vectorization

Vectorization for diagnostic

Fluorescence microscopy

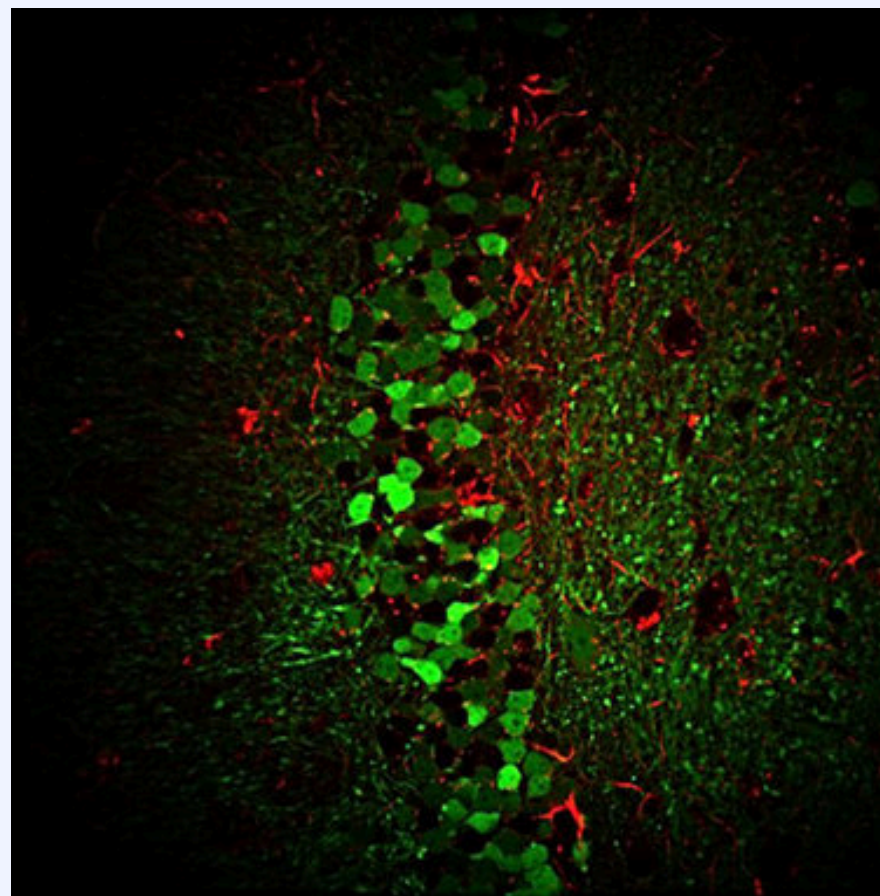


Figure 7: Confocal microscope images from brain sections of mice injected with LV-GFP vector stained with neuronal markers [9].

Nuclear Medicine

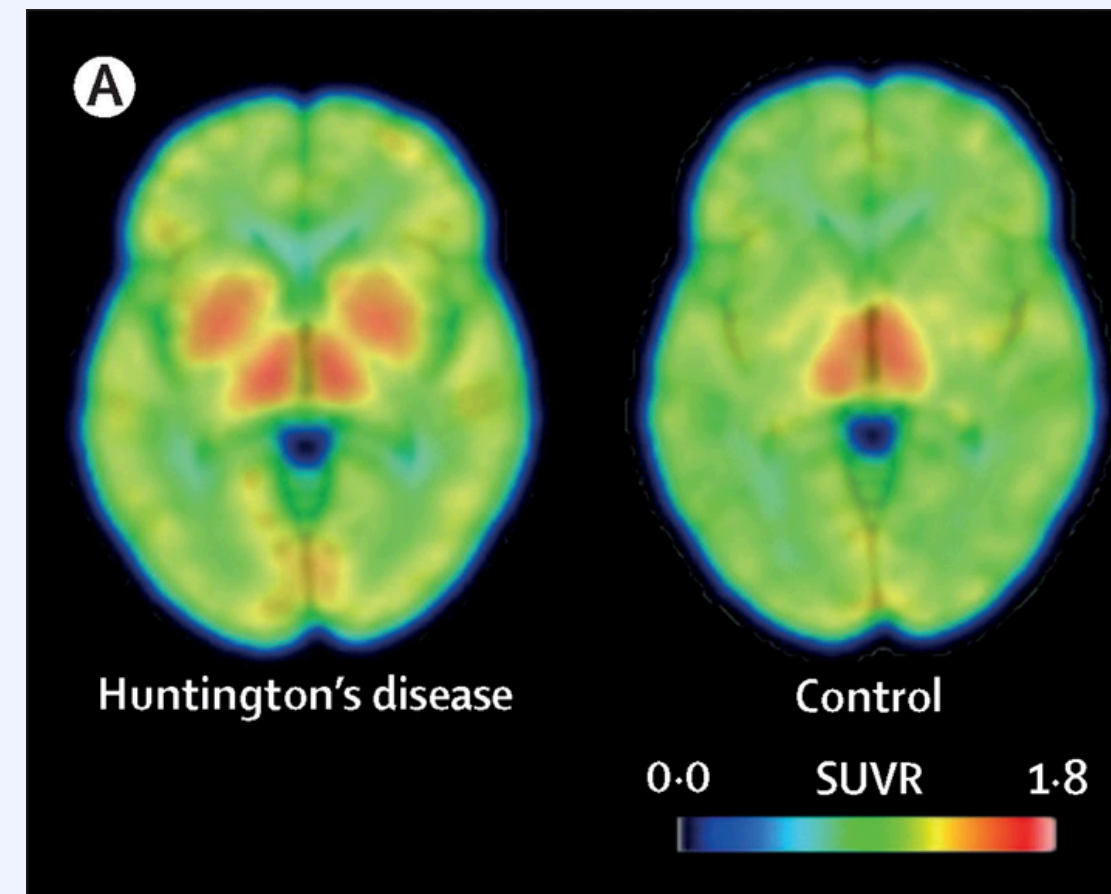


Figure 8: TSPO imaging in Huntington's disease, PET images [10].

Vectorization

Viral Vectors

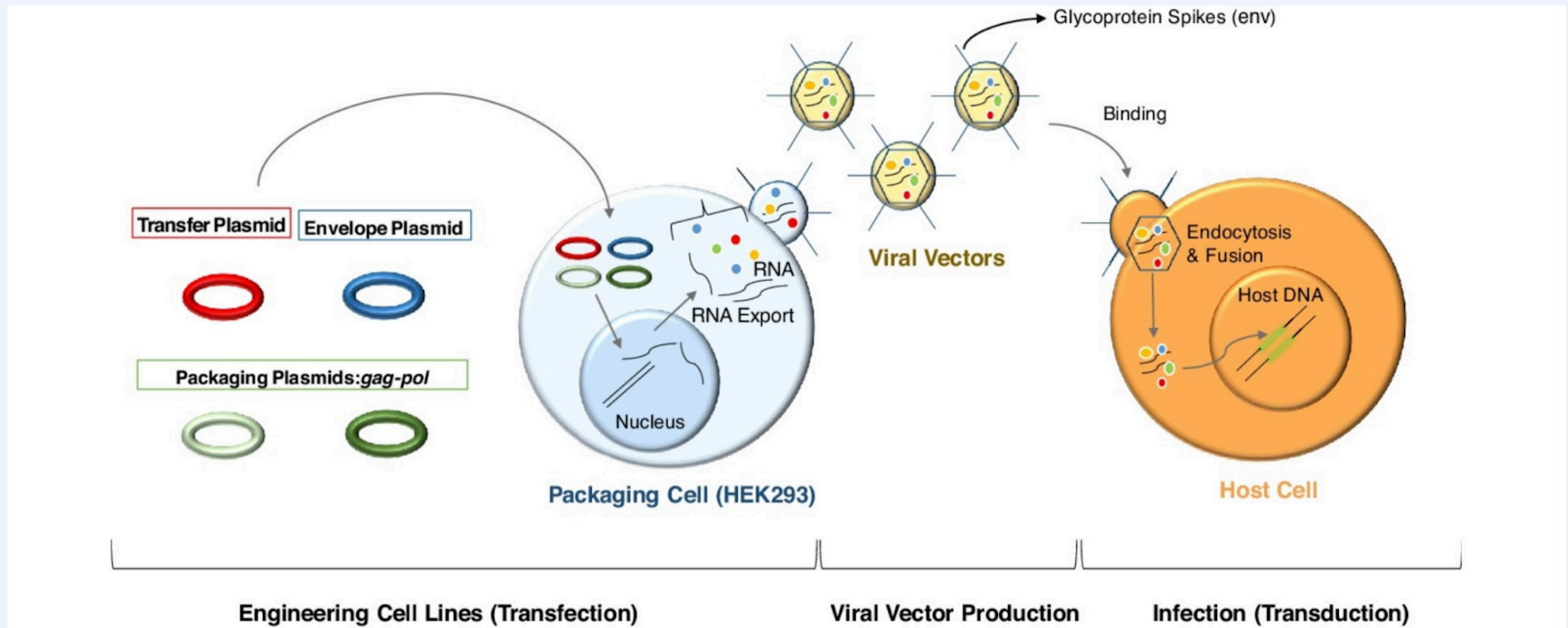
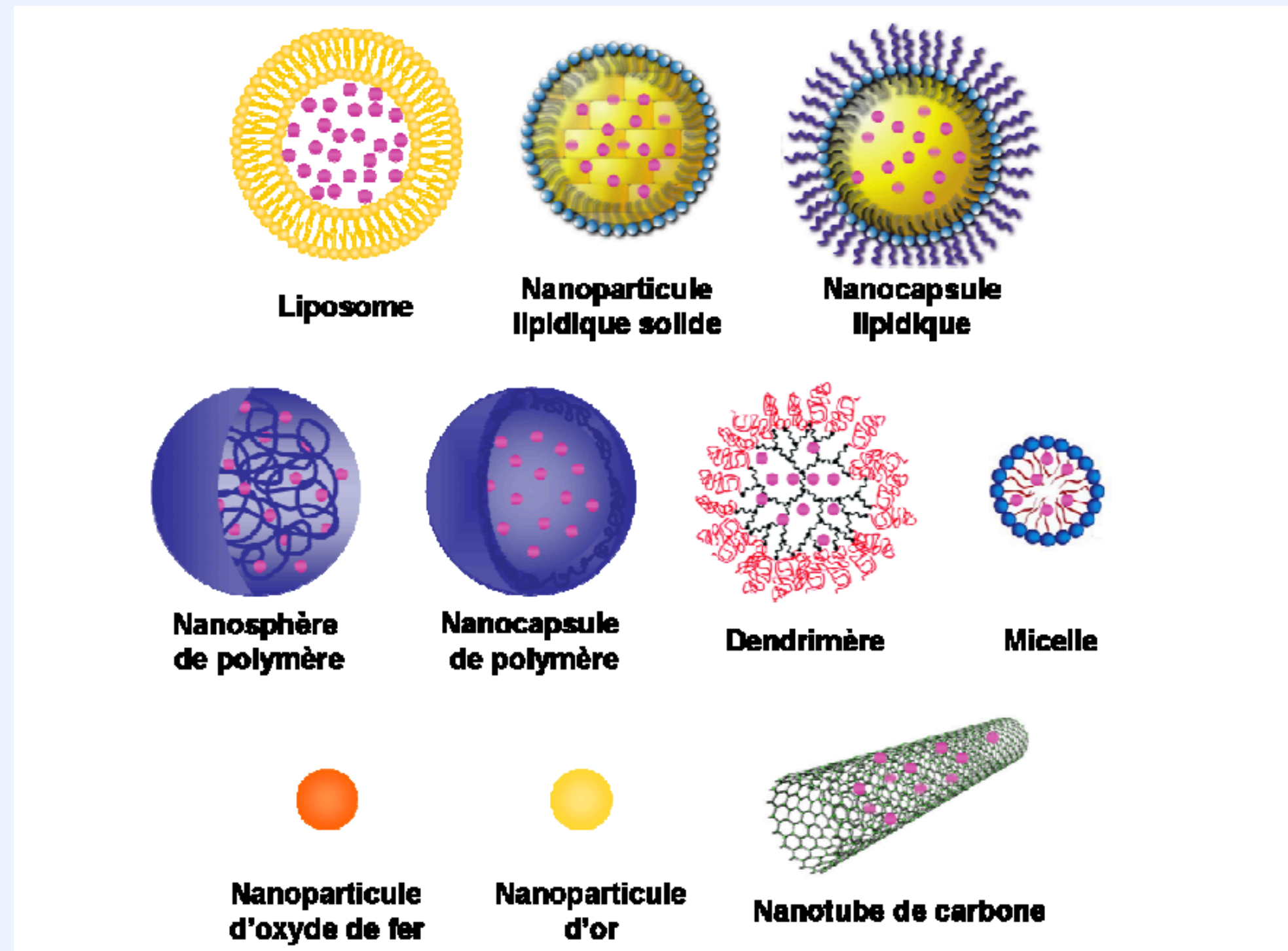


Figure 9: Viral Vector's process through a packaging cell to integrate the host cell. [7].

Vectorization

Non-Viral Vectors



Vectorization

Viral and Non-Viral Vectors's Characteristics

	Types	Advantages	Limitations	Fields of application	Treatment application
Viral Vectors	- Adenoviruses - Adeno-associated viruses - Retroviruses - Lentiviruses - Herpesviruses	- Targeting precision - Delivering Efficiency - Cross biological barriers - Long-term expression - Low immunogenicity - Customizability	- Payload capacity - Immune responses - Potential toxicity - High production complexity - Integration risks	- Immunotherapy - Gene therapy	- Cancer - Respiratories diseases - Huntington's disease - Parkinson's disease - Alzheimer's disease - ...
Non-Viral Vectors	- Lipoplexes - Polyplexes - Dendrimers - Aptamers - Nanoparticles	- Safety - Cost-effectiveness - Flexible cargo capacity - Reduced immune response - Customizability	- Low transfection Efficiency - Short-term expression - Poor Targeting - Stability issues	- Gene therapy - Chemotherapy - Diagnostic : PET scan, SPECT scan, CZT camera	- Cancer - Cardiovascular diseases - Genetic disorders - Cystic Fibrosis - Diabetes - Retinal disorders - ...

Case study:

AAV5-miHTT as therapeutic
vector for Huntington's Disease

Case study

Vector AAV5

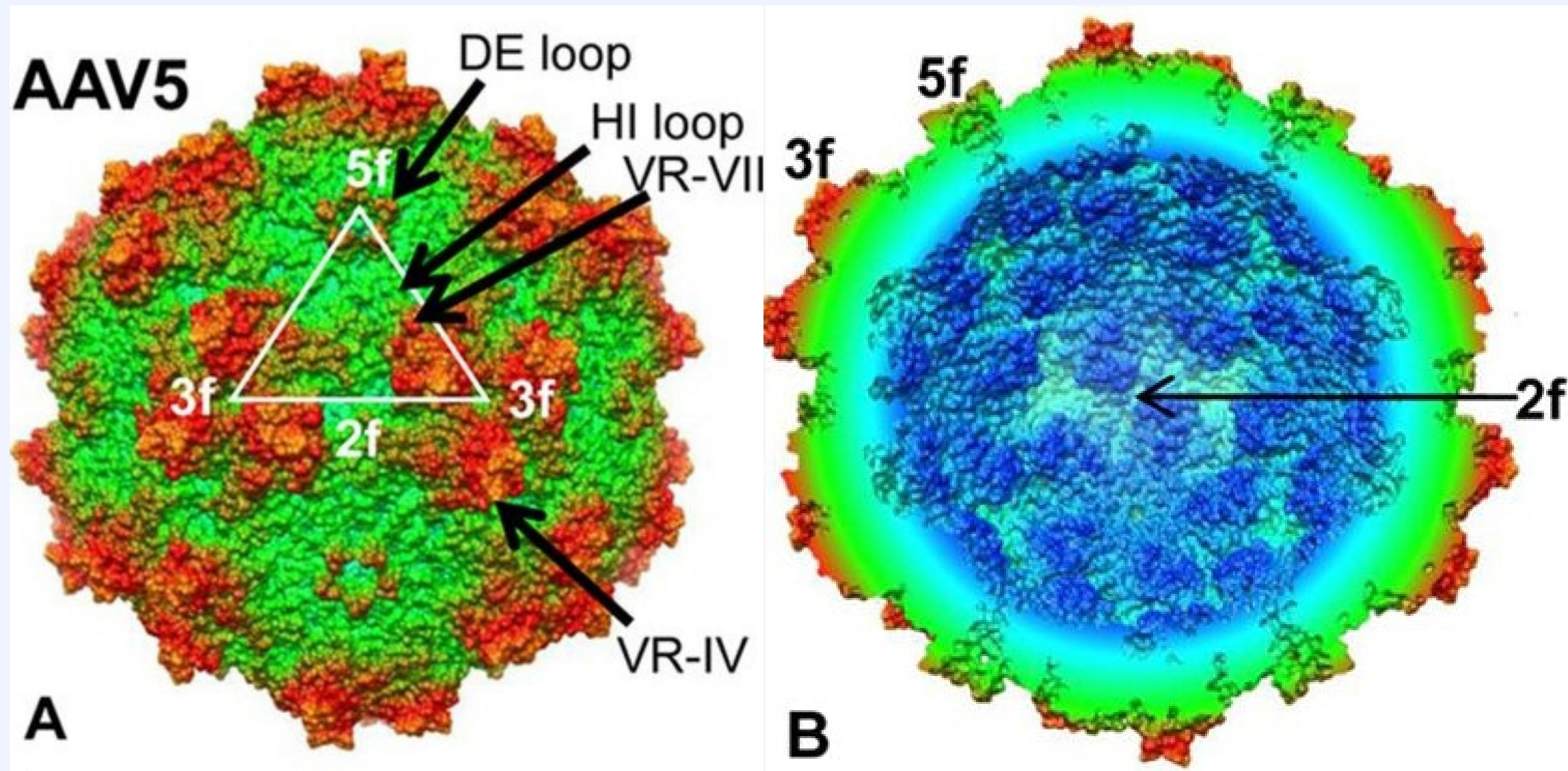


Figure 11: 3D representation of the Adeno-Associated Virus serotype 5 (AAV5) [11].

- Diameter of 20 to 25 nanometers.
- Icosahedral shape (a polyhedron with 20 faces).
- 4.7 kilobases of genetic material

Case study

Therapeutic Payload

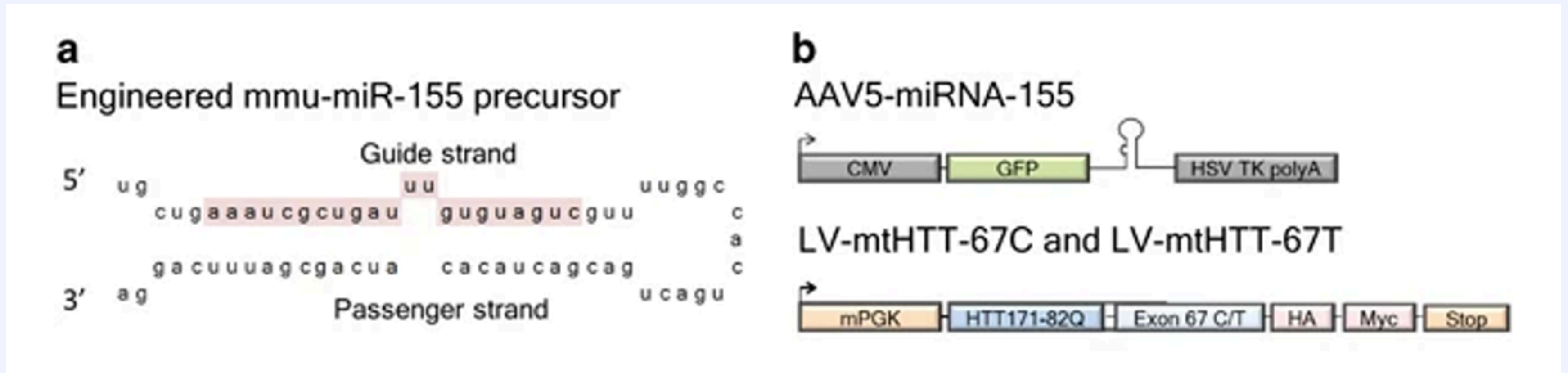


Figure 12: Design of a proof-of-concept study using the AAV5-miHTT-155 and AAV5-miSNP67T-155 vectors in an acute HD rat model. a. The structure and sequence of the engineered mmu-pre-miR-155 precursor. b. Schematic representation of the AAV5-miHTT-155 and AAV5-miSNP67T-155 expression cassettes; and LV-mtHTT-67C and LV-mtHTT-67T expressing the mutant HTT sequences. . [12].

Case study

Delivery strategies

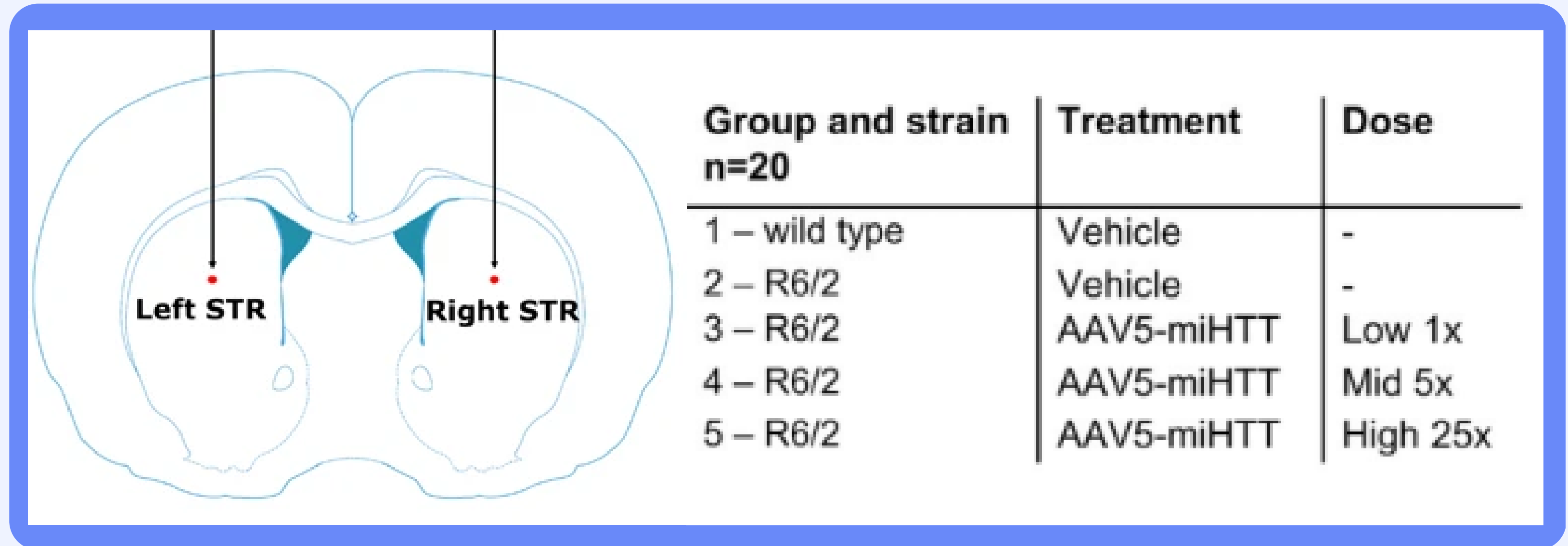


Figure 13: AAV5-miHTT Administration in the Mouse Brain Using Bilateral Injections in the Striatum and Experimental Setup Showing wild-type and Q175 KI mice used in the study with the different dose groups. [13].

Case study

Mechanisms of Therapeutic Action

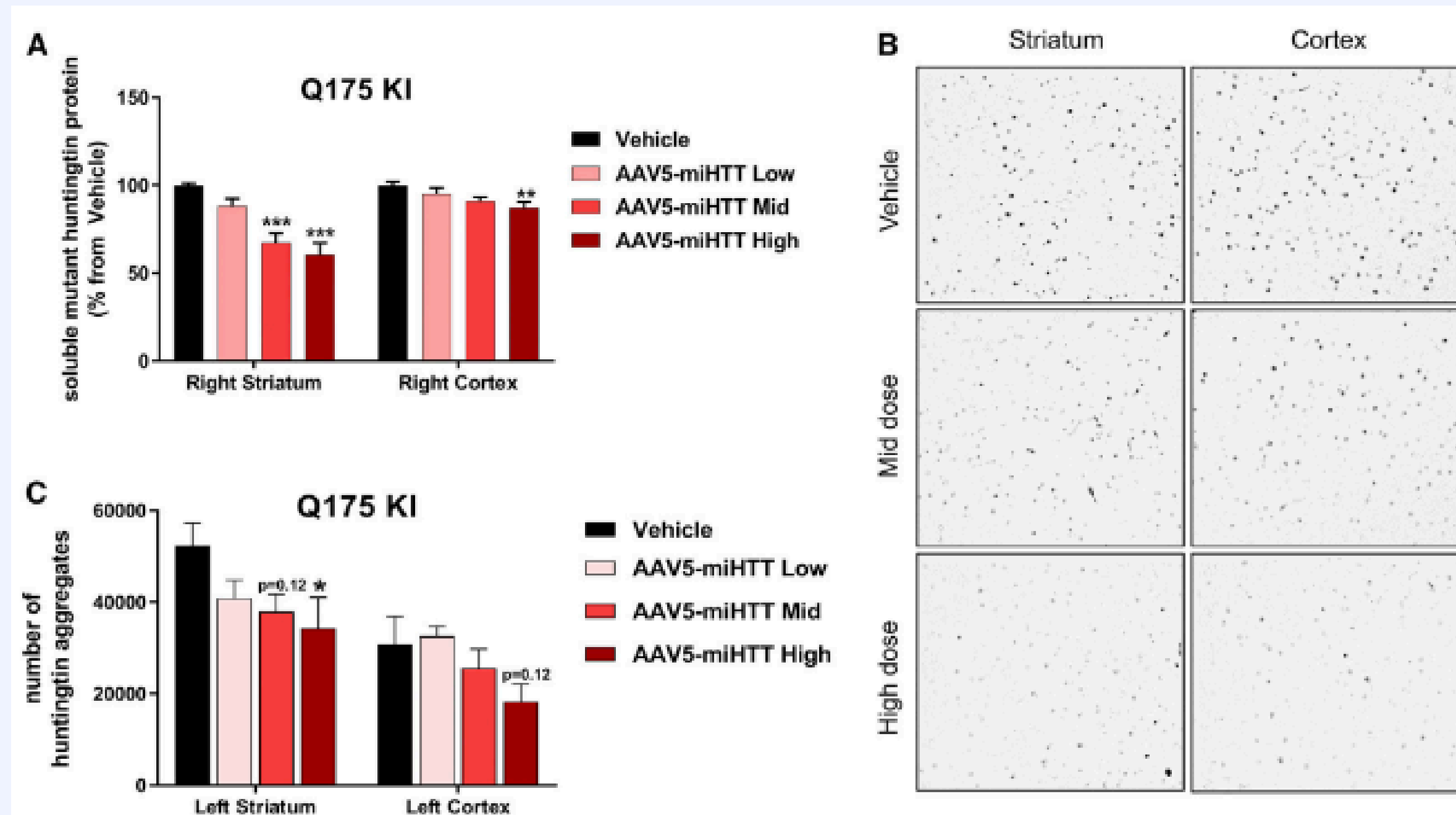


Figure 14: AAV5-miHTT Treatment Demonstrates Human Mutant Huntingtin Target Engagement in Q175 KI Mouse Striatum and Cortex [14].

Case study

Targeted Distribution and Bioavailability.

AAV5-miHTT-155
AAV5-miSNP67T-155

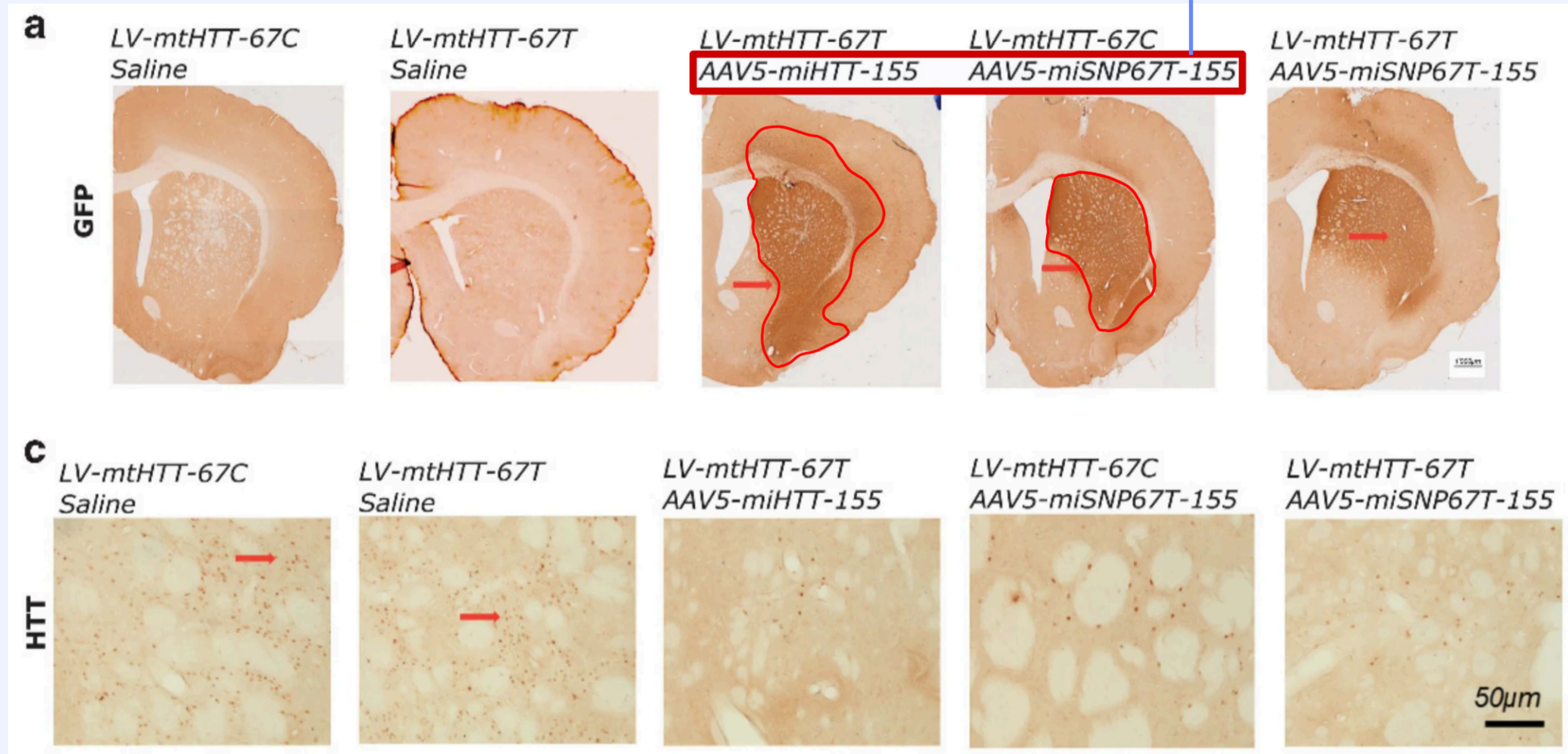


Figure 15: Phenotypic improvement of HD neuropathology following AAV5-miHTT-155 and AAV5-miSNP67T-155 injections in HD rats [15].

Case study

Targeted Distribution and Bioavailability

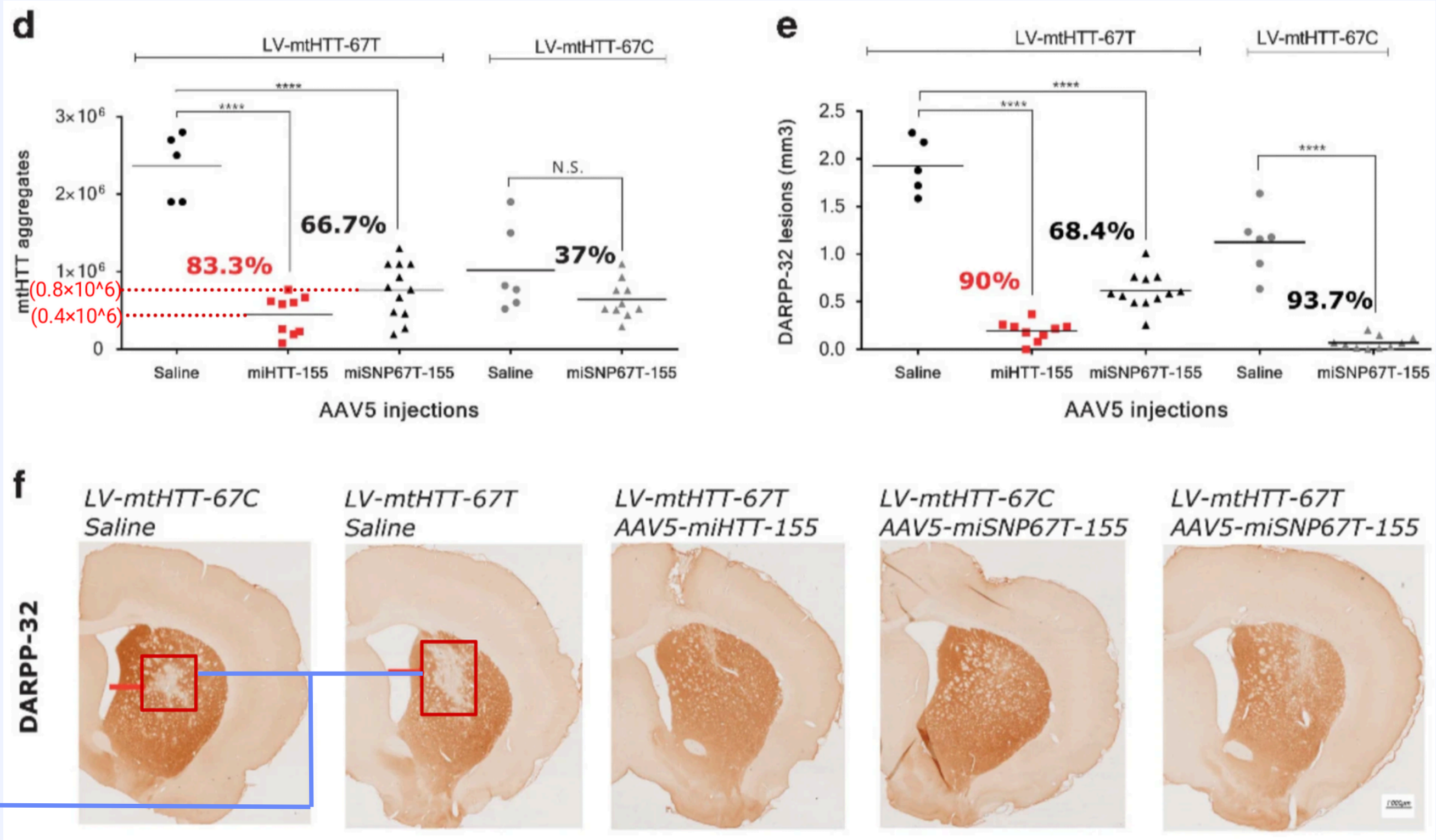


Figure 15: Phenotypic improvement of HD neuropathology following AAV5-miHTT-155 and AAV5-miSNP67T-155 injections in HD rats [15].

Case study

Preclinical Validation

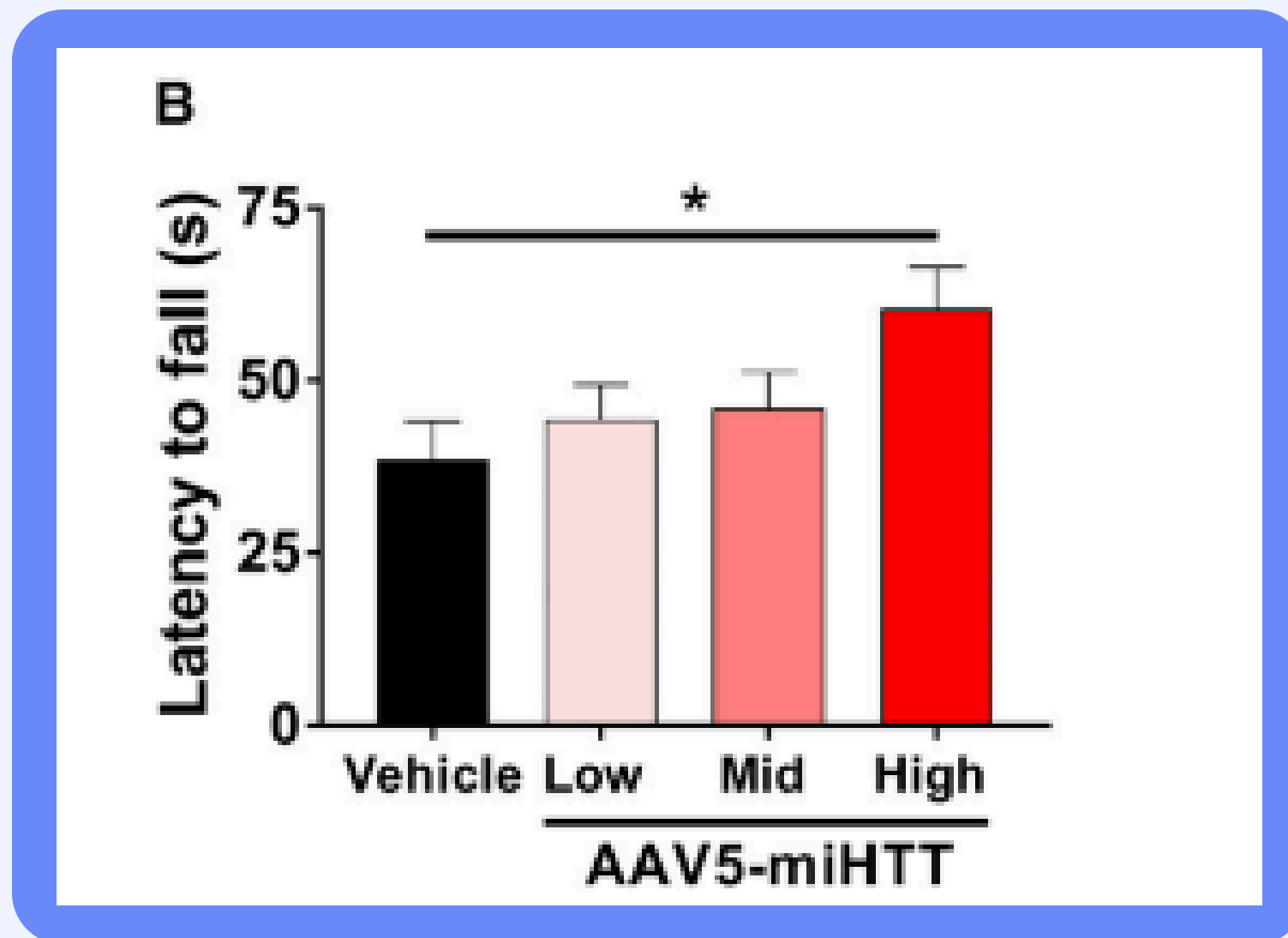


Figure 16: AAV5-miHTT Treatment Shows Motor Coordination Improvement in R6/2 Mice [16].

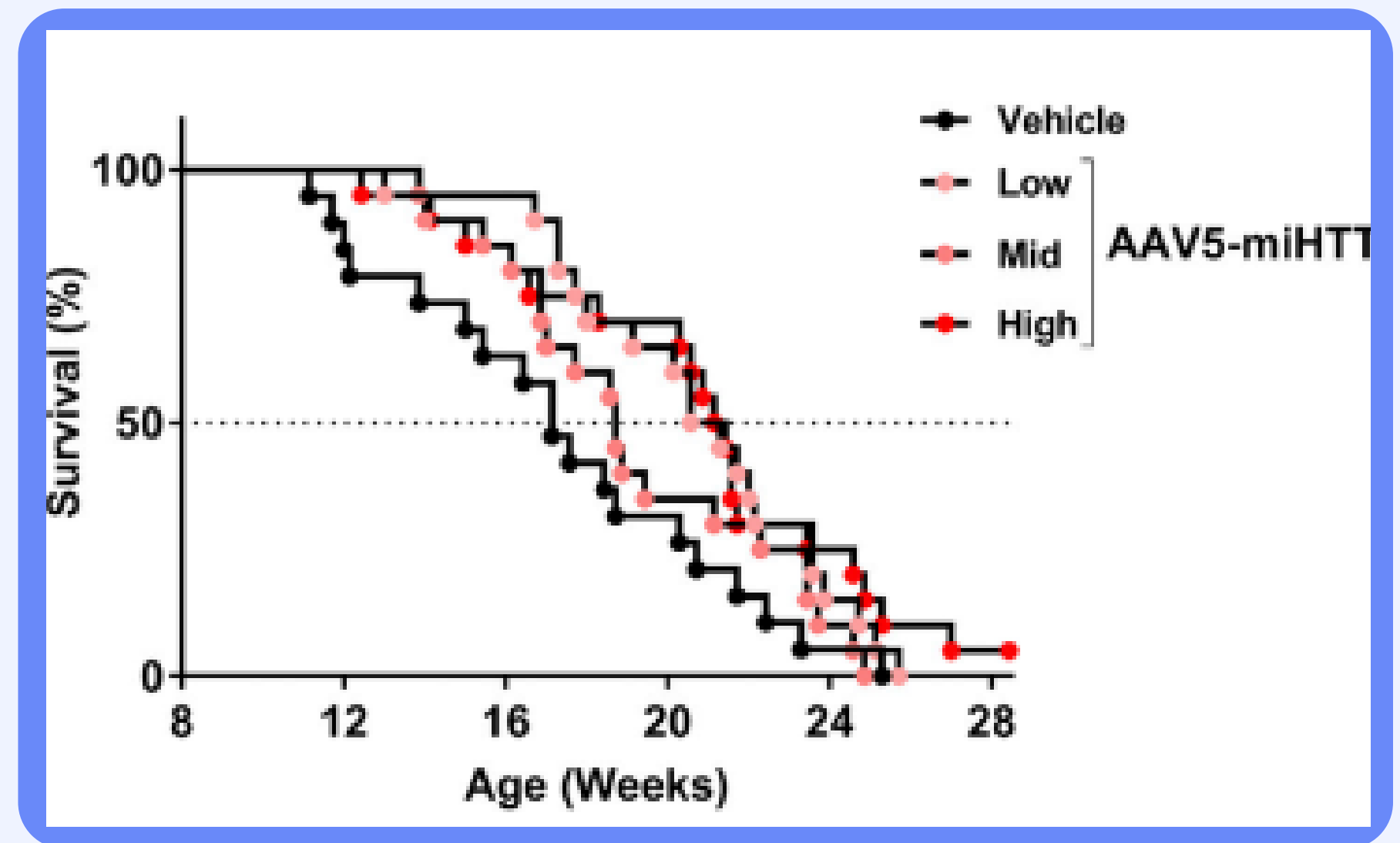


Figure 17: Kaplan-Meier Curve Showing Survival of Untreated and AAV5-miHTT-Treated R6/2 Mice [17].



Conclusion



Thank you for your attention

Questions?

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